

Bayesian Black-Box Optimisation

Project FAQ

Author: Erkan Keskin - Neuroxa-Labs

https://github.com/Neuroxa-Labs/Bayesian_Optimisation

Companion to docs/final_report.pdf

Version: v1 (pre-Week-13 portal results)

Imperial College London - PCMLAI Stage 2 - Neuroxa-Labs

Imperial College London - PCMLAI Stage 2 capstone Author: Erkan Keskin - Organisation: Neuroxa-Labs
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This FAQ answers common questions about this repository and how the weekly black-box challenge was run. It is a portfolio companion to the final research report, not a substitute for the official course FAQ / brief.

A. Challenge basics

A1. What is the task?

Maximise eight unknown continuous black-box functions $f_i : [0,1]^{d_i} \rightarrow \mathbb{R}$ (dimensions 2 through 8). Each week you may submit one query per function. The portal returns a scalar y . Higher y is always better. The true formulas are withheld.

A2. What counts as a "query"?

A six-decimal portal string such as 0.440000-0.980000-0.980000-0.980000. Coordinates must lie in $[0,1]$. After the portal returns y , that pair is appended to the function's history and used for the next decision.

A3. Do we need to recover the formula?

No. The goal is sample-efficient sequential decisions that find high-performing x values under a tiny weekly budget - not symbolic identification.

A4. How many observations do we have?

Each function starts with an initial design (seed points in `data/function_/_`). Weekly portal points are appended. After Week 12, typical totals are roughly 22-52 points depending on dimension and seed size (see the final report table). Function 3 currently has 11 weekly rows in `data/` (no fabricated Week-12 point).

B. Method

B1. What is the primary optimiser?

A Gaussian Process with a Matérn kernel and ARD length scales, scored with Expected Improvement (EI) or Upper Confidence Bound (UCB), plus a trust region around the incumbent for late exploit.

B2. Why not a neural net as the weekly decider?

With $n \sim 20$ -50 points, calibrated uncertainty matters more than flexible curve-fitting. Neural nets / large ensembles were used by some peers as diagnostics; they were not our primary portal decider.

B3. What is the F1 "trust gate"?

Most of the F1 map reads ~ 0 (near-null labels). Exploiting a GP on a null map wastes budget on boundary artefacts. The trust gate refuses aggressive exploit until a measurable signal cluster appears; Weeks 10-12 opened a lobe near (0.64, 0.68).

B4. Why hard-return on F2 and F6?

Both have sharp basins. A "nearby" neighbour step can collapse y (F2 ~ 0.54 vs peak ~ 0.78 ; F6 Week-11 collapse after leaving the Week-10 centroid). Late policy returns toward the historical best after a failed neighbour.

B5. What changed in Weeks 10-13?

Near-pure exploitation of proven regions: shrink trust regions, move only ARD-sensitive axes, hard-return after misses, and (F5) continue the ridge climb on x_1 .

C. Results (after Week 12)

C1. What are the best-so-far values?

- Fn	Best y	Status note
- F1 | 7.711×10^{-16} | Unresolved absolute peak; late signal lobe
- F2 | 0.776645 | Strong sharp ridge
- F3 | -0.011366 | Safe local band
- F4 | 0.678600 | Strong late climb (W12)
- F5 | 3800.74 | Clearest ridge success
- F6 | -0.136 | W10 basin; fragile
- F7 | 1.872233 | Strong late climb (W12)
- F8 | 9.872928 | Strong late climb (W12)

C2. Did late weeks still improve?

Yes. Improve counts: W8 3/8 - W9 4/8 - W10 5/8 - W11 4/8 - W12 4/8 (F4, F5, F7, F8).

C3. Is F1 "solved"?

Not in this repository. Seed max remains the official best. Peers who found a large positive peak (e.g. ~ 0.5 -2) demonstrate that a true spike exists; our Weeks 10-12 lobe is the first usable local basin we observed.

C4. When will Week 13 results appear?

Week 13 queries are locked in weeks/WEEK13_STRATEGY.md. Portal y values are pending; append with scripts/append_week13.py and update the final report table when they arrive.

D. Repository map

D1. Where is the executable code?

notebooks/BBO_Capstone_Optimized.ipynb (../notebooks/BBO_Capstone_Optimized.ipynb)

D2. Where is the written evidence pack?

- Markdown: docs/final_report.md (final_report.md)
- PDF: docs/final_report.pdf (final_report.pdf)
- This FAQ (PDF): docs/project_faq.pdf (project_faq.pdf)

D3. Where are figures and progress dashboards?

- Gallery: reports/analysis/README.md (../reports/analysis/README.md)
- Progress: reports/progress/README.md (../reports/progress/README.md)
- Per-function plots: data/function_/analysis_F.png

D4. Where are transparency artefacts?

DATASHEET.md (../DATASHEET.md) - MODEL_CARD.md (../MODEL_CARD.md) -
docs/TECHNICAL_JUSTIFICATION.md (TECHNICAL_JUSTIFICATION.md)

D5. How do I navigate course-style deliverables?

docs/COURSE_INDEX.md (COURSE_INDEX.md)

E. Practical / reproducibility

E1. Can I reproduce a weekly decision?

Yes in spirit: reload that week's history from data/function_/, fit the GP with the notebook settings, and re-score candidates. Exact floating-point portal strings are logged in weekly strategy notes under weeks/.

E2. Are random seeds fixed?

GP multi-restart fits and candidate searches can retain stochasticity. Treat weekly notes + locked portal strings as the audit trail for what was actually submitted.

E3. What should I read first as a new visitor?

- Root README.md (./README.md)
- This FAQ or final_report.pdf (final_report.pdf)
- reports/analysis/README.md (./reports/analysis/README.md) for visuals

F. Honest limitations

- One query per function per week - no inner real-function line search.
 - GPs can be confidently wrong in empty regions (inflated ? -> boundary chase).
 - Clustered sampling can miss a distant second mode (especially F1).
 - F2's observed peak may be an optimistic draw under noise.
 - F3 weekly count in data/ is 11; we did not fabricate a missing row.
- Version: v1 (pre-Week-13 portal results). Update C1 / C4 when Week 13 y values arrive.